

Pratik Bairoliya

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EXPERIENCE

Capital One

Jul. 2026 – Present

Software Engineer II

McLean, VA

- Own the nightly **end-of-day** run that posts transactions, settles balances, and closes the ledger for millions of Card, Bank, and Financial Services accounts.
- Lead **SSUI** integration into the end-of-day APIs, moving account servicing off legacy vendor systems without missing a nightly cycle.
- Stand up observability for a new **Kubernetes + Temporal** orchestration platform, surfacing failed cycles to on-call before downstream teams feel them.
- Ship that platform through **Argo CD** and **Helm** across environments, keeping every release one revision from rollback.

Capital One

May 2024 – Jul. 2026

Software Engineer I

McLean, VA

- Took a **batch data persistence API** from zero to production in **Go** — the system of record for every nightly cycle — secured with **PoP token** auth.
- Designed a **block-allocation account ID service** that reserves IDs in blocks instead of one call at a time, letting Card and Bank open accounts in bulk without contending for a shared sequence.
- Built the nightly **Databricks** job that aggregates Card end-of-day runs and reconciles totals against the ledger, catching breaks before downstream teams see them.
- Automated nightly publication of **80M+ records** across **2M+ accounts** to downstream data platforms via end-to-end performance testing and **Hive** sort tuning on **AWS EMR**.
- Rebuilt the military rewards catalog off SharePoint as a **Vue.js / Spring Boot / DynamoDB** app, **2.8×** faster (\$9M projected revenue), deployed with **Terraform**, **Docker**, and **Jenkins**.

Fidelity Investments

Jun. 2023 – May 2024

Software Engineer

Durham, NC

- Migrated **45%** of mainframe stock and job requests to **AWS** using **Terraform**, serving **400+** internal associates.
- Modeled **10,000** mainframe metadata nodes in **Neo4j** and wrote **Cypher** analysis that removed **5,000+** duplicates.
- Delivered the full stack on a 4-person Agile team — **Angular** front end, **Java** services, **S3** metadata sync.

PROJECTS

Daily Lookback | *Python, Ollama (local LLM), Google APIs, SQLite, launchd*

2026

- Daily briefing generator pulling Calendar, Gmail, and iMessage into one note — **100% on-device**.
- Wrapped nondeterministic LLM output in a deterministic rules layer and append-only state, so scheduled reruns are **idempotent**; fully stubbed end-to-end tests.

Receipt Pipeline | *Python, Apple Vision OCR, Ollama, file watchers*

2026

- Scan-to-structured-data pipeline: OCRs any receipt, classifies line items with a local LLM, emits spend reports.
- Fingerprint dedup over an append-only index as source of truth; **52 tests**, file-locked against concurrent runs.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript/JavaScript, SQL, Cypher, Bash, C/C++

Cloud & Infrastructure: AWS (EMR, S3, DynamoDB, EC2), Kubernetes, Temporal, Docker, Terraform, Argo CD, Helm, Jenkins

Backend & Data: REST API design, distributed batch processing, Hive, Databricks, Spring Boot, PostgreSQL, Neo4j

Frontend: Vue.js, React, Angular, HTML/CSS

AI/ML: Local LLM inference (Ollama), prompt design, OCR pipelines, PyTorch

Practices: Distributed systems design, observability, performance testing, zero-downtime migration, Agile

Certification: AWS Certified Solutions Architect – Associate (Mar. 2026)

EDUCATION

North Carolina State University

Aug. 2021 – May 2025

B.S. Computer Science and B.S. Applied Mathematics — GPA: 4.0/4.0

Raleigh, NC

- Park Scholar** — full merit scholarship awarded to the top **1%** of applicants for leadership and service.

LEADERSHIP & INTERESTS

Leadership: Chairman of InspireNC (2020–2025), a student-run nonprofit bringing FIRST Robotics mentorship to NC schools.

Interests: Underwater robotics, running the Krispy Kreme Challenge, and home-lab automation.